

ESP32-S3 OTA Update Guide - Command Line Method

Prerequisites

- ESP32 already running with OTA code
 - ESP32 connected to WiFi
 - Know the ESP32's IP address
-

Step-by-Step OTA Update Process

Step 1: Make Your Code Changes

1. Open your sketch in Arduino IDE
2. Make whatever changes you want (code updates, bug fixes, new features)
3. Save the sketch (**Ctrl+S** or **File** → **Save**)

Step 2: Export Binary File

1. In Arduino IDE, go to **Sketch** → **Export Compiled Binary**
2. Wait for compilation (bottom of IDE will show progress)
3. When you see "**Done compiling**", the .bin file is ready
4. The .bin file is saved in your sketch folder

Step 3: Locate Your .bin File

Option A: Show Sketch Folder

1. Go to **Sketch** ▢ **Show Sketch Folder**
2. Navigate to: build\esp32.esp32.esp32s3\
3. Find the file: YourSketchName.ino.bin
4. Right-click ▢ **Copy as path**

Option B: Find it via PowerShell

```
Get-ChildItem -Path "$env:USERPROFILE\Documents\Arduino" -Filter "*.bin" -Recurse | Where-Object  
{$_LastWriteTime -gt (Get-Date).AddMinutes(-10)} | Select-Object FullName
```

Step 4: Get ESP32's IP Address

If you don't know the IP:

1. Connect ESP32 via USB
2. Open **Serial Monitor** (Ctrl+Shift+M)
3. Press **RESET** button on ESP32
4. Look for: IP address: 192.168.1.XX
5. Write down this IP address

Or check your router's connected devices list

Step 5: Upload via Command Line

Open **PowerShell** (Windows) or **Terminal** (Mac/Linux) and run:

Windows:

```
python "C:\Users\Shree\AppData\Local\Arduino15\packages\esp32\hardware\esp32\3.3.4\tools\esptota.py" -i  
YOUR_ESP32_IP -p 3232 --auth=admin -f "PATH_TO_YOUR_BIN_FILE"
```

Replace:

- YOUR_ESP32_IP ▢ Your ESP32's IP (e.g., 192.168.1.16)

- `PATH_TO_YOUR_BIN_FILE` = Full path to your .bin file
- `admin` = Your OTA password if you changed it

Example:

```
python "C:\Users\Shree\AppData\Local\Arduino15\packages\esp32\hardware\esp32\3.3.4\tools\esptool.py" -i
192.168.1.16 -p 3232 --auth=admin -f "C:\Users\Shree\Documents\Arduino\OTA_test\build
\esp32.esp32.esp32s3\OTA_test.ino.bin"
```

Step 6: Wait for Upload to Complete

You should see:

```
Uploading.....
[=====] 100%
Done
```

RGB LED Status:

- **Purple blinking** = Upload in progress
- **Green** = Upload successful
- **Red** = Error occurred

The ESP32 will automatically restart with the new firmware!

Quick Reference Template

Save this as a text file and just replace the values each time:

```
# My OTA Upload Command Template
python "C:\Users\Shree\AppData\Local\Arduino15\packages\esp32\hardware\esp32\3.3.4\tools\esptool.py" -i
192.168.1.16 -p 3232 --auth=admin -f "C:\Users\Shree\Documents\Arduino\YOUR_SKETCH\build
```

\esp32.esp32.esp32s3\YOUR_SKETCH.ino.bin"

Just update:

- IP address (if it changes)
 - Sketch name
 - Password (if changed)
-

Pro Tips

Tip 1: Create a Batch File (Windows)

Create a file called upload_ota.bat in your sketch folder:

```
@echo off
set ESP_IP=192.168.1.16
set PASSWORD=admin
set BIN_FILE=%~dp0build\esp32.esp32.esp32s3\%~n0.ino.bin

echo Uploading to %ESP_IP%...
python "%LOCALAPPDATA%\Arduino15\packages\esp32\hardware\esp32\3.3.4\tools\espota.py" -i %ESP_IP% -p
3232 --auth=%PASSWORD% -f "%BIN_FILE%"
pause
```

Then just:

1. Export binary in Arduino IDE
2. Double-click upload_ota.bat
3. Done!

Tip 2: Find espota.py Path (if different version)

If you have a different ESP32 package version, find espota.py:

```
Get-ChildItem -Path "$env:LOCALAPPDATA\Arduino15\packages\esp32" -Filter "espota.py" -Recurse | Select-Object  
FullName
```

Copy the path shown and use it in your upload command.

Tip 3: Verify Before Upload

Before uploading, ping the ESP32 to make sure it's online:

```
ping 192.168.1.16
```

If you get replies, it's ready for OTA.

Tip 4: Check Upload History

To see your recently compiled binaries:

```
Get-ChildItem -Path "$env:USERPROFILE\Documents\Arduino\*\build\*\*.bin" | Sort-Object LastWriteTime -  
Descending | Select-Object FullName, LastWriteTime -First 5
```

Common Errors & Solutions

Error: "No response from device"

Solution: ESP32 is offline or IP changed

- Ping the IP to verify connection
- Check Serial Monitor for current IP
- Restart ESP32 and try again

Error: "Authentication failed"

Solution: Wrong password

- Check the password in your code: `ArduinoOTA.setPassword("your_password");`
- Update the `--auth=` parameter to match

Error: "File not found"

Solution: Wrong .bin file path

- Make sure you exported the binary first
- Check the path has no typos
- Use quotes around paths with spaces

Error: "Update error"

Solution: Not enough space or corrupted file

- Re-export the binary
- Check partition scheme supports OTA
- Try uploading via USB, then try OTA again

Workflow Summary

Every time you want to update:

1. ✎ Edit code in Arduino IDE
2. ✎ Save changes

3. ▢ Sketch ▢ Export Compiled Binary
4. ▢ Copy .bin file path
5. ▢ Open PowerShell
6. ▢ Run espota.py command with your IP and .bin path
7. ▢ Wait for "Done"
8. ▢ New code is running!

Time: ~30 seconds per update once you get the hang of it!